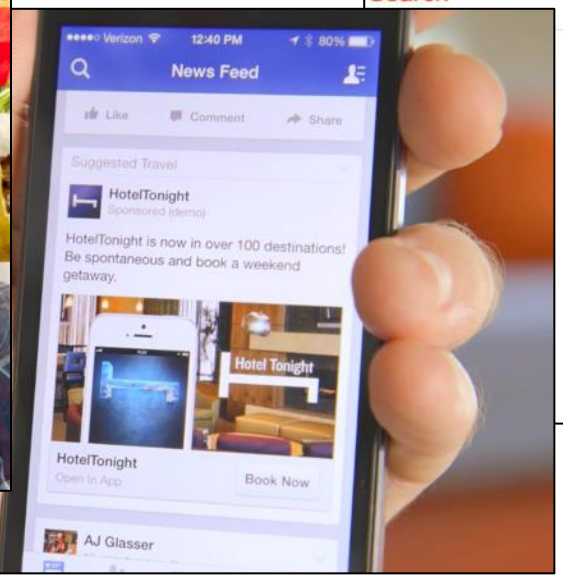
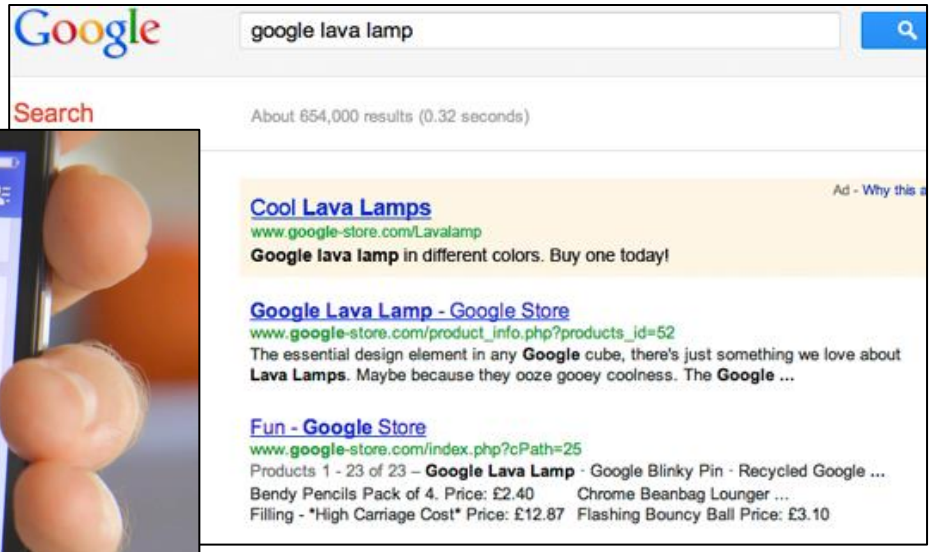
Ads and Auctions

Slides by Raed Almomani and Andrew DeOrio

Motivation

- Ads are the business model behind most of the web
- Google sponsored links, Facebook, Snapchat, Instagram, YouTube, etc





Selling ads before the web

- Newspaper ads sold by size and placement
  - Bigger ads cost more
  - Different sections cost different $
- TV ads sold by time slot and show popularity
  - Different shows and different time of day cost different $
  - Upfront season – a few months before fall when new shows are announced and advertisers can buy air time

Selling ads on the Web

<u>Before the Web:</u>

- Content known in advance
- Ada bought ahead of time

<u>On the Web:</u>

- Content not known in advance
- Ads are sold “just in time”
  - E.g., before every search on Google!
- Use automated *auctions* to select the ad and price

Pay per click

<u>Before the Web:</u>

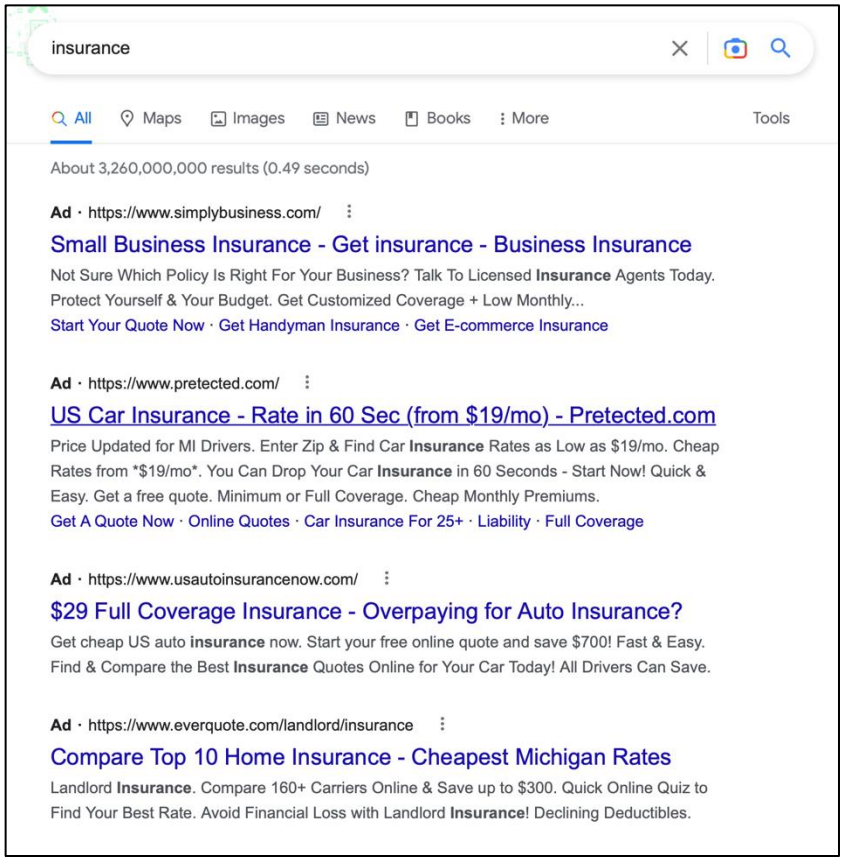
- Estimated audience size
- Number of newspaper subscribers, TV viewers, etc.

<u>On the Web:</u>

- Measure audience more accurately
- Record when a user clicks on the ad
- *Pay per click*

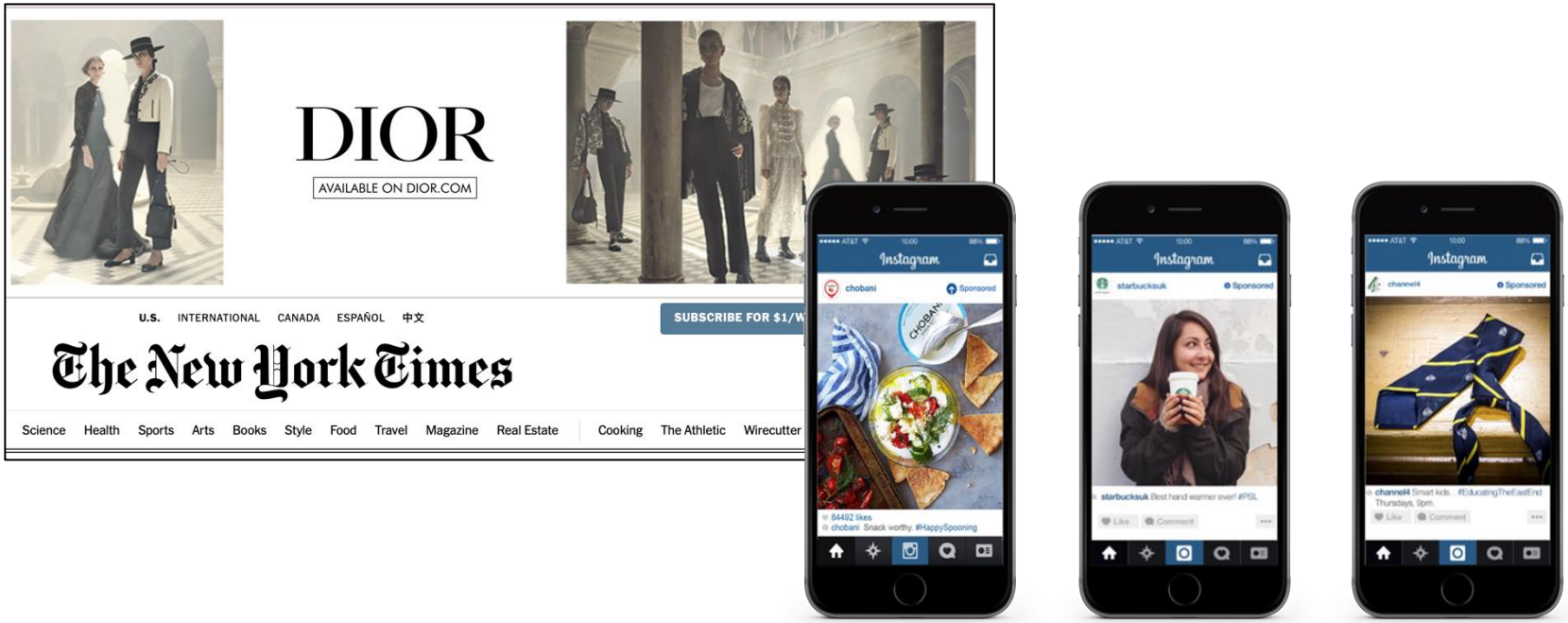
Search engine advertising

- Search engine advertising
  - AKA search engine marketing
- Pay to be the top link



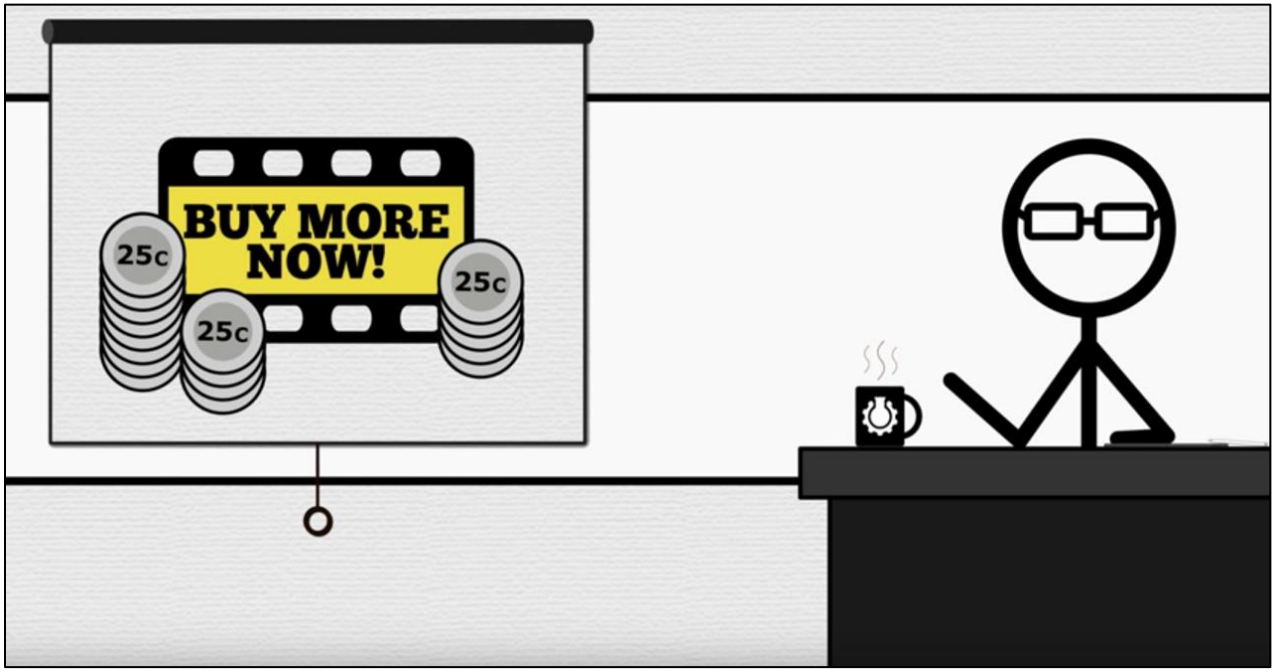
Display advertising

- Display advertising
  - AKA Banner ads
  - Common in websites, apps, and social feeds
  - Can include images, text, or video



YouTube ads in action

- https://www.youtube.com/watch?v=KW0eUrUiyxo



Agenda

- **Auctions in the real world**
- Auction types and terminology
- Auction mathematical analysis
- Auctions for selling ads

Real world auctions in action

- https://www.youtube.com/watch?v=ACKT5jWJHTI
- Cattle auction in Kansas
- Bidding on the price in dollars per 100lb live weight

Real world auction observations

- Auctions help many participants discover a price
- No single actor needs to manually set that price
- The same idea make auctions useful for web

Real world auction observations

- Expensive to run an auction and get the attention of multiple potential buyers in a short window
- Used for
  - Commodities, livestock
  - Art, antiques, rare items
- Auctions become much cheaper to run online
  - Particularly if there is no human in loop

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Auction mechanisms

- Different auctions rules lead to different outcomes
- Designing auction rules is called ***Mechanism design***
- *Players* AKA *agents* in an auction have different motives and strategies
- *Game theory* helps us predict bidder behaviour

The players

- A seller
- Multiple *bidders*
- An item being sold
- Each bidder has a private valuation
- Each bidder wants to win while paying as little as possible

The players: who knows what

- *Seller* does not know the *bidders*' valuations of the *thing*
- *Bidders* do not know each others' valuations
- Yet the auction must still determine a price

Many kinds of auctions

- *Open auction*: Everybody sees the bids
  - Ascending-price ("English" auction)
  - Descending-price ("Dutch" auction)

- *Sealed-bid auction*: Bids are secret
  - First-price
  - Second-price (AKA "Vickrey" auction)

English auctions

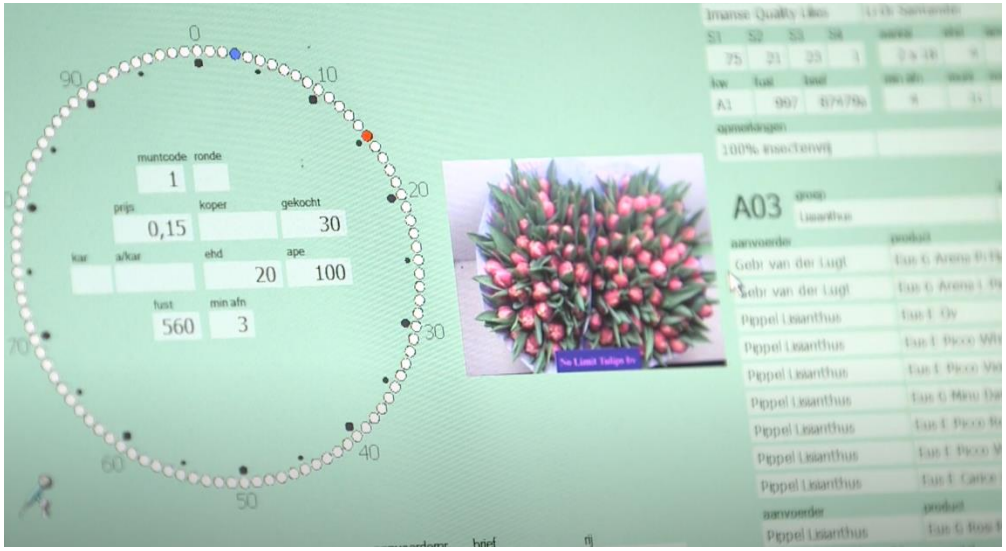
- Open, ascending-price auction
- Price goes up and up, until only one bidder left
- Winner pays the final standing price
- Best-known type

Dutch auctions

- Open, descending price auction
- Price starts high and drops, until someone accepts
- Common in flower auctions

Dutch auctions

- Dutch Flower Auction
- https://www.youtube.com/watch?v=FUJoRMD0B-Y&t=160s



Sealed bid auctions

- Sealed-bid, first-price
  - Everyone bids silently, at once
  - Highest bid wins, and pays own bid
- Sealed-bid, second-price ("Vickrey")
  - Everyone bids silently, at once
  - Highest bid wins, pays 2nd-highest bid

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Analysis

- Open, ascending-price (English)
  - Who wins? What does the winner pay?
  - What is the bidder's likely strategy?
- Sealed-bid second-price auction (Vickrey)
  - Who wins? What does the winner pay?
  - What is the bidder's likely strategy?

Analysis

- Open, ascending-price (English)
  - Who wins? What does the winner pay?
    - Bidder with highest value, pays about 2nd-highest value
  - What is the bidder's likely strategy?
    - Bidder stays in until price is higher than bidder's value
- Sealed-bid second-price auction (Vickrey)
  - Who wins? What does the winner pay?
    - Bidder with highest value, pays 2nd-highest value
  - What is the bidder's likely strategy?
    - Bidder bids true value
- In both cases, winner is the bidder with highest value, pays 2nd-highest value
  - Technically, English auction winner pays 2nd-highest + minimum bid increment, assuming optimal strategy

Sealed-bid, second-price analysis

- Sealed-Bid, 2nd-Price auctions (Vickrey)
- How much should you bid?
- True value? More? Less?

Sealed-bid, second-price analysis

- If you bid above your value, you risk overpaying
- If you bid below your value, your risk losing when you should have won
- Therefor, bidding your **true value** is best

Sealed-bid, second-price analysis

- Bid does not determine the *price you pay*
- Bid determines *whether you get to pay*

- 2nd-price auction bidders are "truthful”

- Same with an open/ascending (English) auction
  - Assuming you follow the optimal strategy

Proxy bidding

- eBay uses **proxy bidding;** Open auction
- 1st-place bidder pays 2nd-place price

- You submit a maximum bid
- The system automatically bids just enough to keep you ahead

Proxy bidding example

- Ru Paul the seller lists a backpack for sale on eBay with a minimum starting bid of $1 and a 5 day duration

Proxy bidding example

Day 1

Current (public) bid: n/a; Minimum next bid: $1; Max: n/a.

- The first bid is from Jack, who bids $10 even though the minimum is $1. Since there are no other bidders, eBay bids the minimum of $1 on Jack's behalf and changes the minimum for any new bids to $1.25—the current bid of $1 plus the increment of $0.25.

Proxy bidding exan

Day 2

Current bid: $1; Minimum next bid: $1.25; Max: Jack, up to $10.

- The second bid is from Jill, who bids $1.25. Because Jack's $10 maximum was higher, eBay immediately bids $1.50 on Jack's behalf ($1.25 plus one increment) and resets the minimum to $1.75 ($1.50 plus one increment).

Proxy bidding exan

Day 3

- **Current bid: $1.50; Minimum next bid: $1.75; Max: Jack, up to $10.**
- Jill tries again, bidding $1.75. Jack's maximum is still higher, so eBay immediately bids on his behalf again and adjusts the minimum accordingly.

Proxy bidding exan

Day 4

- **Current bid: $2; Minimum next bid: $2.25; Max: Jack, up to $10.**
- Frustrated, Jill tries a fourth time, bidding $16.50. This is higher than Jack's maximum, so eBay sets the current bid to $10.50 (Jack's maximum plus one increment). Jill is now the high bidder and eBay is bidding automatically on her behalf.

Proxy bidding exan

Day 5

- **Current bid: $10.50; Minimum next bid: $11; Max: Jill, up to $16.50.**
- During the final moments of the auction, a new bidder named Tim bids the minimum of $11. eBay automatically bids on Jill's behalf, outbidding Tim by one increment at $11.50.

Proxy bidding exan

End

- **Winning bid: $11.50; Winner: Jill.**
- The auction ends after exactly five days. Jill is the highest bidder and wins, *not* for $16.50 (her maximum bid), but for $11.50 (one increment higher than her nearest competitor's bid).

Sniping

- Sniping uses software to bid
- Sniping places a bid at the last possible moment; avoid giving competitors time to respond.
- What would happen if everyone sniped?

Sniping analysis

- What would happen if everyone sniped?
- If everyone waits until the end, bids become effectively sealed.
- Sniping effectively turns a Proxy Bidding auction into a Vickrey Auction, because it "seals" the bids

Sealed-bid, first-price analysis

- Sealed-Bid, 1st-Price Analysis
- What is the bidder's likely strategy?

Sealed-bid, first-price analysis

- Sealed-Bid, 1st-Price Analysis
- What is the bidder's likely strategy?
  - In a first-price auction, your bid determines your payment.
  - Bidding your full value may leave little or no benefit
  - So bidders usually bid below their true value.

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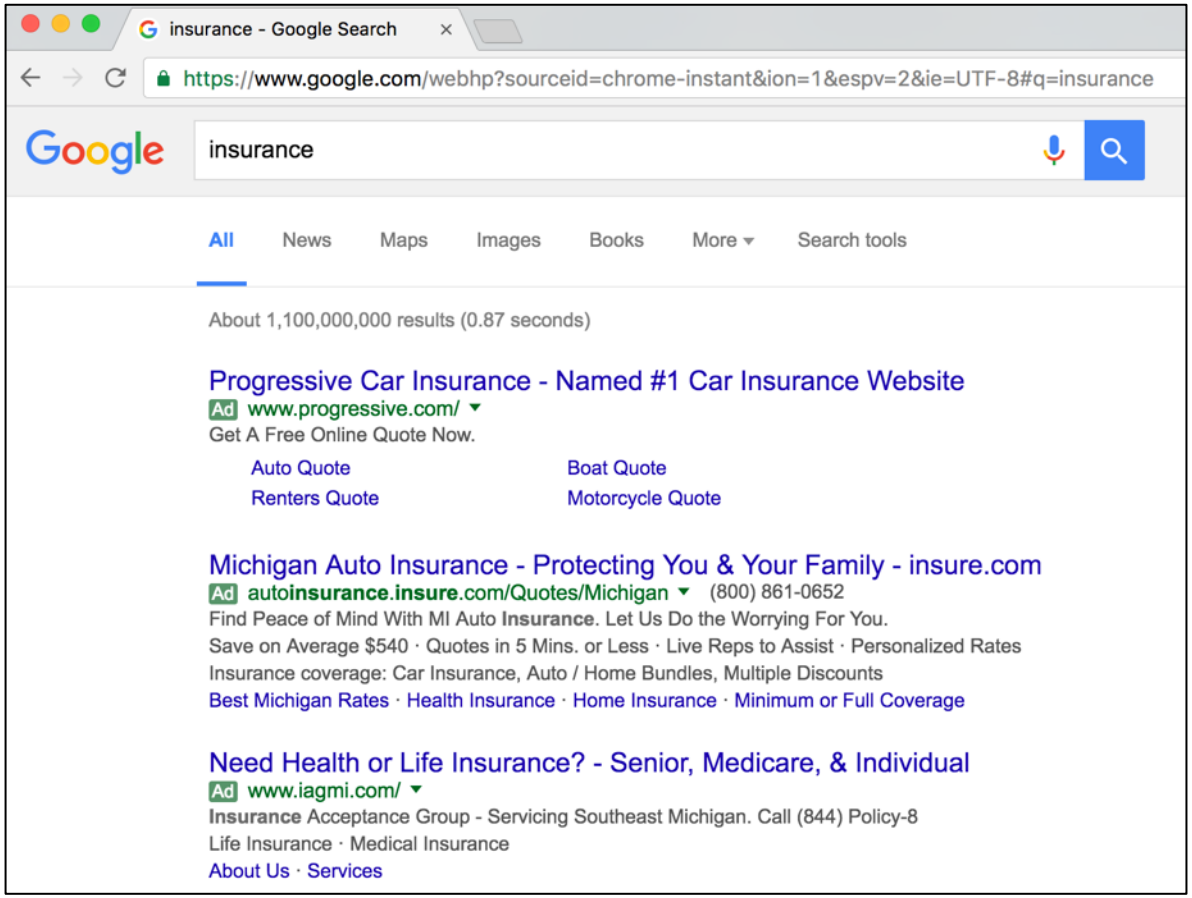
Pay-per-click (PPC) auctions

- Auctions determine which ads you see



Overture pay-per-click auctions

- Overture was the first company to do text search ads, even before Google!
  - Bought by Yahoo in 2003
- Bidders purchase "keywords"
  - Ads displayed in order, highest first
- If the user clicks the ad, high bidder pays bid price
  - User can click on many, or no, ads
- Bids are visible to other bidders
- **First-price, open auction**

Overture example

- Three parties bid on the keyword "eecs485"

$1

$2

$3

Overture example

- R2D2 is pos #1
  - He bids $3, pays $3 upon user click
- Neytiri is pos #2
  - She bids $2, pays $2 upon user click
- ET is pos #3
  - He bids $1, pays $1 upon user click

- What should R2D2 do?

Overture example

- R2D2 is pos #1
  - He bids $3, pays $3 upon user click
- Neytiri is pos #2
  - She bids $2, pays $2 upon user click
- ET is pos #3
  - He bids $1, pays $1 upon user click

- What should R2D2 do?
- Decrease bid to $2.01, and save $0.99 per click

Overture example

- R2D2 is pos #1
  - **He bids $2.01, pays $2.01 upon user click**
- Neytiri is pos #2
  - She bids $2, pays $2 upon user click
- ET is pos #3
  - He bids $1, pays $1 upon user click

- What should Neytiri do?
- Increase bid to $2.02: with just $0.02 more per click, she can get first position!!

Overture example

- **Neytiri is pos #1**
  - **She bids $2.02, pays $2.02 upon user click**
- R2D2 is pos #2
  - He bids $2.01, pays $2.01 upon user click
- ET is pos #3
  - He bids $1, pays $1 upon user click

- What happens next?

Overture example

- It’s advantageous to change your bid rapidly
  - Pay enough to be on top, but no more
- Has some bad social effects: lots of time/money spent on bidding

Google Ads for search

- A few critical differences:
  - Sealed-bid
  - 2nd-price payment
  - Competitors' bids are hidden

Google Ads example

- Neytiri is pos #1
  - She bids $3.00, but pays $2.01 upon click

- R2D2 is pos #2
  - He bids $2.00, pays $1.01 upon click
- ET is pos #3
  - He bids $1, pays minimum
- Great! But…

Google Ads example

- Unlike Overture, Google auction prices are kept secret. You don't know what other people bid.
- But you can guess. What if R2D2 changes his bid to $2.75?

Google Ads example

- Neytiri is pos #1
  - She bids $3.00, but pays $2.76 upon click.
  - Her cost increased due to R2D2!

- R2D2 is pos #2
  - He bids $2.75, still pays $1.01 upon click

- ET is pos #3
  - He bids $1, pays minimum

Google Ads example

- More differences:
  - Displayed ads ranked depends on both:
    - Bid amount
    - Ad-quality
  - Click-through rates (CTR): how often people click your ad

- Google was originally the "cheap competitor" to Overture, then overtook

Google Ads caveats

- Google AdWords is not exactly Vickrey auction
  - Standard Vickrey auctions sell one item
  - Google sells multiple items
  - *This is called* ***Generalized Second Price*** auction

- What if you want two items, but if you can't get them both, you want neither?
  - E.g., cotton candy machine + 32 tons of raw cotton candy

Summary

- Auctions are automated on the web
- Primarily used to sell ads
  - Sellers with ad platforms like Google, Facebook, etc.
  - Buyers who want to advertise
- Enable just-in-time ad sales customized to users